

Big Data Analytics
CISELEC 501A-CS41S1

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Lab Activity-03: CRISP-DM | Business Understanding (TOTAL: 60 pts)

OBJECTIVE: This activity is designed to explore the initial phase of the CRISP-DM framework—Business Understanding—through scenario-based analysis. It involves identifying real-world problems across various industries, locating relevant datasets, and examining the structure and format of extracted data. The exercise emphasizes the importance of aligning data mining efforts with clearly defined business goals, resource considerations, and contextual constraints, forming a foundation for informed decision-making and effective analytical planning.

Instructions:

1. Define the business problem in your own words (5 pts per scenario).
 2. Extract a dataset from the suggested source.
 3. Answer the following questions:
 - How was the data extracted? (3 pts per scenario)
 - What is the format of the data (CSV, Excel, PDF)? (2 pts per scenario)
 - What is the composition of the data (number of records and fields)? (5 pts per scenario)
-

Scenario 1: Retail Inventory Optimization

- **Industry:** Retail
- **Business Problem:** A retail chain wants to predict which products will sell more during holiday seasons to avoid overstocking or understocking.
- **Suggested Dataset:** >> [Click Here](#) <<

Answers:

1. The business problem is where a retail chain is wondering and wants to find out what products sell more during the holiday season. They don't want any products that could lead to overstocking or understocking. Thus, wanting to predict which product they should focus on.

```
import pandas as pd
dataset_link = "https://raw.githubusercontent.com/AI-Artisans/bda_cs41s1/refs/heads/main/20250823-Lab-Wk5/retail_store_inventory.csv"
df = pd.read_csv(dataset_link)
print(df.head())
```

	Date	Store ID	Product ID	Category	Region	Inventory Level
0	2022-01-01	S001	P0001	Groceries	North	231
1	2022-01-01	S001	P0002	Toys	South	204
2	2022-01-01	S001	P0003	Toys	West	102
3	2022-01-01	S001	P0004	Toys	North	469
4	2022-01-01	S001	P0005	Electronics	East	166

	Units Sold	Units Ordered	Demand Forecast	Price	Discount
0	127	55	135.47	33.50	20
1	150	66	144.04	63.01	20
2	65	51	74.02	27.99	10
3	61	164	62.10	32.72	10
4	14	135	9.26	73.64	0

	Weather Condition	Holiday/Promotion	Competitor	Pricing	Seasonality
0	Rainy	0	0	29.69	Autumn
1	Sunny	0	0	66.16	Autumn
2	Sunny	1	1	31.32	Summer
3	Cloudy	1	1	34.74	Autumn
4	Sunny	0	0	68.95	Summer

2.

3. Ff questions:

- Data was extracted by reading the raw file. We can use pandas library from python and using its global method “read_csv”
- format of the data is in a form of **csv format**.
- Composition of the dataset contains products id, category, units ordered, units sold, etc.. which are crucial for our business problem.

Scenario 2: Hospital Readmission Analysis

- **Industry:** Healthcare
- **Business Problem:** A hospital wants to reduce readmission rates by identifying common factors among patients readmitted within 30 days.
- **Suggested Dataset:** >> [Click Here](#) <<

Answers:

1. The problem of a certain hospital is an increase of readmission rates amongst patients that are readmitted within 30 days. It is needed to identify common factors amongst these patients so we can reduce the readmission rates.

```
import pandas as pd
dataset_link = "https://raw.githubusercontent.com/AI-Artisans/bda_cs41s1/refs/heads/main/20250823-Lab-Wk5/diabetic_data.csv"
df = pd.read_csv(dataset_link)
print(df.head())
```

	encounter_id	patient_nbr	race	gender	age	weight
0	2278302	8222157	Caucasian	Female	[0-10]	?
1	149190	55629189	Caucasian	Female	[10-20]	?
2	64410	86047875	AfricanAmerican	Female	[20-30]	?
3	500364	82442376	Caucasian	Male	[30-40]	?
4	16680	42519267	Caucasian	Male	[40-50]	?

	admission_type_id	discharge_disposition_id	admission_source_id
0	6	25	1
1	1	1	7
2	1	1	7
3	1	1	7
4	1	1	7

	time_in_hospital	... citoglipton	insulin	glyburide-metformin
0	1	...	No	No
1	3	...	No	Up
2	2	...	No	No
3	2	...	No	Up
4	1	...	No	Steady

	glipizide-metformin	glimepiride-pioglitazone	metformin-rosiglitazone
0	No	No	No
1	No	No	No
2	No	No	No
3	No	No	No
4	No	No	No

	metformin-pioglitazone	change diabetesMed	readmitted
0	No	No	No
1	No	Ch	Yes
2	No	No	Yes
3	No	Ch	Yes
4	No	Ch	Yes

[5 rows x 50 columns]

2.

3. Ff questions:

- Data was extracted by reading the raw file. We can use pandas library from python and using its global method “read_csv”

- format of the data is in a form of **csv format**.
- Composition of the dataset contains encounter_id, admission_type_id, readmitted, etc.. which are crucial for our business problem.

Scenario 3: Local Airport Delay Prediction

- **Industry:** Transportation
- **Business Problem:** A particular Airport wants to improve flight scheduling by analyzing delay patterns based on historical flight data.
- **Suggested Dataset:** >> [Click Here](#) <<

Answers:

1. In a particular airport, there exists a problem in which the flight schedule is inefficient. There are historical data that contains timelines of flight schedules. By analyzing patterns of the schedule delay, it can help improve airport's flight schedules.

```
import pandas as pd
dataset_link = "https://www.transtats.bts.gov/Marketing_Annual.aspx?heY_fryrp6lrm4=FDPI&heY_fryrp6Z106u=I&heY_gvzr=%20E&heY_fryrp6v10=E"
tables = pd.read_html(dataset_link)
df = tables[5]
print(df.head())
```

Found 6 tables

	Year	OnTime	Arrivals	OnTime (%)	Arrival Delays	Delayed (%)	\
0	2018	2538228	70.05	564525	17.78		
1	2019	2562232	78.27	621622	18.99		
2	2020	1960372	78.15	262281	10.46		
3	2021	1900120	86.74	251176	11.47		
4	2022	2184577	76.45	572797	20.05		

	Flight Cancelled	Cancelled (%)	Diverted	Flight Operations
0	64855	2.04	7300	3174908
1	81336	2.48	8217	3273406
2	281896	11.24	4034	2508583
3	35171	1.61	4288	2190675
4	93586	3.28	6510	2857471

- 2.
3. Ff questions:
 - Data was extracted by reading the html tags of the url. We can use the pandas library from python and use its global method "read_html" and take the position(5 -> 6th index) of the table that we want.
 - format of the data is in a form of **html table**.
 - Composition of the dataset contains OnTime Arrivals, Arrival Delays, Year, Flight Operations, etc.. which are crucial for our business problem to solve and identify patterns.

REFLECTION:

- **Why this data matters to the business? (5 pts)**

These datas matters to our business as it indicates timelines and histories that were being recorded for that particular business. Using those recorded information can lead us to identify different problems that we can think of in order to help improve or solve underlying problems to the business.

- **What risks or constraints might affect the project (e.g., missing data, privacy concerns)? (5 pts)**

Risks or constraints that affect and can have a huge impact on a particular project are those data that are inconsistent, values that scales more than or higher the algorithm that you chose can compute, datas that are too noisy that you are unable to use and such other factors.

- **How their findings could influence decision-making? (5 pts)**

Their findings could influence decision making by showing a definitive proof in a form of computation and algorithms that shows a scientific proof that this output really has something to do with the problem that they needed to decide on.

**" I affirm that I have not given or received any unauthorized help in
this assignment, and that this work is my own "**

Kenneth Arias

